Welcome

to Week Two SQL & AI Bootcamp

28/03/2026

Be so good that the Industry can't Ignore you!

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AGENDA

PART 1 — WHAT IS DATA?

PART 2 — OPERATIONAL DATABASES

PART 3 — HOW DATA IS STRUCTURED

PART 4 — THE DATA WAREHOUSE

PART 5 — DATA MODELLING IN PRACTICE USING AI

PART 6 — SQL: THE LANGUAGE OF DATA

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PART 1 — WHAT IS DATA?

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What is data, and what is a database?

Data = any recorded information (names, dates, clicks, transactions)

Database = organised system for storing and retrieving data

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Data is generated every time something happens.

Every interaction with a digital system — a tap, a click, a purchase — creates a record. Understanding how that data flows from event to storage is key to understanding databases.

Common data sources

User actions

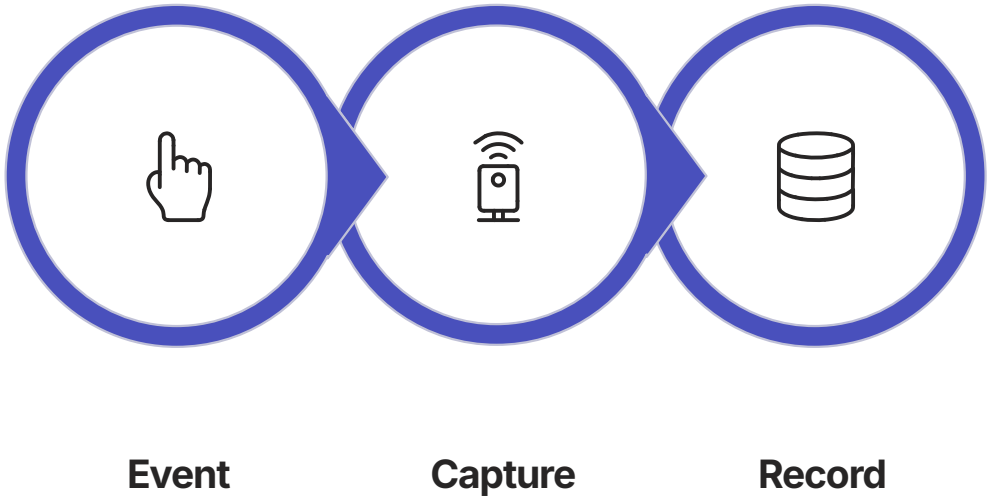
Clicks, form submissions, logins, purchases

Physical events

Card taps, sensor readings, device signals

System events

Timestamps, errors, background transactions



Every digital interaction follows this same pattern: **Event → Capture → Record.**

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Data in memory — the temporary home

- RAM is the working surface.
- Disk is the permanent record.

RAM	Disk
Temporary working space	Permanent storage
Where work actually happens	Where data lives long-term
Extremely fast	Slow to read and write
Expensive per gigabyte	Cheap per gigabyte
Gone the moment power cuts	Survives power cuts

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More Memory Examples in the next Tab

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PART 2 — OPERATIONAL DATABASES

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Making data generated permanent

- Called Operational databases
- Also called transactional databases
- Designed to handle many users reading and writing simultaneously
- Data written to disk — survives crashes, restarts, power cuts
- Powers: banking apps, booking systems, hospital records, e-commerce

User action → Application → Operational Database (on disk) → Data persisted ✓

Three-column layout for the types of database:

Relational	NoSQL	In-Memory
Tables, rows, columns	Flexible formats	Lives in RAM
SQL Server, PostgreSQL	MongoDB, Redis	Redis, Memcached
Banking, healthcare	Social media, IoT	Caching, real-time

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ACID — four rules that keep data trustworthy.

A — Atomicity	C — Consistency
All or nothing. No halfway transactions.	Rules are never broken. Valid in, valid out.

I — Isolation	D — Durability
Concurrent transactions can't interfere.	Confirmed = permanent. Survives crashes.

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PART 3 — HOW DATA IS STRUCTURED

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HOW DATA IS STRUCTURED

The flat approach, and why it causes problems

Let's say a school app stores student data like this:

StudentName	Course	TeacherName	TeacherEmail	Grade
Helen Jones	SQL Basics	Dr. Patel	d.patel@school.ac.uk	A
Helen Jones	Python Intro	Dr. Patel	d.patel@school.ac.uk	B
James Wright	SQL Basics	Dr. Patel	d.patel@school.ac.uk	C

- 🔴 Duplication — same teacher info repeated on every row
- 🔴 Update risk — change email in one place, miss it in another
- 🔴 Deletion problem — delete last student, lose the course
- 🔴 Insertion problem — can't add a teacher without a fake student

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Normalisation

Every fact lives in exactly one place.

The principle (show prominently):

Split data into focused tables. Link them with references. No duplication.

Students

StudentID	StudentName
S001	Amara Jones
S002	James Wright
S003	Priya Sharma

Teachers

TeacherID	TeacherName	TeacherEmail
T01	Dr. Patel	d.patel@school.ac.uk
T02	Ms. Okafor	b.okafor@school.ac.uk
T03	Mr. Chen	w.chen@school.ac.uk
T04	Dr. Mensah	a.mensah@school.ac.uk

StudentID

TeacherID

Enrolments

StudentID	CourseID	Grade
S001	C01	A
S001	C02	B
S002	C01	C
S003	C03	A
S002	C04	B

CourseID

Courses

CourseID	CourseName	TeacherID
C01	SQL Basics	T01
C02	Python Intro	T02
C03	Data Modelling	T01
C04	Cloud Basics	T03

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More Normalization Examples in the next Tab

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PART 4 — THE DATA WAREHOUSE

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Why businesses need a data warehouse

Operational databases run the business. The warehouse understands it.

The problem (show side by side):

Operational DB	Data Warehouse
Built for transactions	Built for analysis
Fast individual reads/writes	Fast complex queries
Current data	Historical data
Don't touch during reporting	Built for reporting

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How data gets into the Data warehouse

Three ways to move data into the warehouse.

Batch (ETL)	Real-time streaming	CDC / Mirroring
Run on a schedule (e.g. nightly)	Continuous, seconds of delay	Watches for changes, replicates them
Simpler to build	Complex, powerful	Elegant, near real-time
Data is hours old	Data is seconds old	Data is near-instant

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Difference between data structures in Operational and Data warehouse

The warehouse is shaped for questions, not transactions.

OLTP vs OLAP comparison:

OLTP	OLAP
Many users, small operations	Few users, large complex queries
Normalised tables	Star schema / denormalised
Operational database	Data warehouse
"Save this appointment"	"How many appointments per month?"

Star schema to illustrate:

- Centre: Fact_Appointments
- Surrounding: dim_Patient, dim_Consultant, dim_Date, dim_Location

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PART 5 — DATA MODELLING IN PRACTICE USING AI

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STAGES OF DATA MODELLING

Data modelling has three stages — each one builds on the last. Three-stage flow:

Data modelling has three stages — each one builds on the last. Three-stage flow:

1. Conceptual
   - Plain language
   - What exists and how it relates
   - No technical detail
   - "Patients have appointments"
2. Logical
   a. Entities and attributes defined
   b. Relationships made precise
   c. ER diagram produced
   d. "Patient has PatientID, DOB, NHSNumber"
3. Physical
   a. Actual tables and columns
   b. Data types and constraints
   c. Specific to your database
   d. PatientDOB DATE NOT NULL

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What bad modelling looks like, and why it matters

A bad model doesn't just slow things down — it makes the data untrustworthy.

Side-by-side comparison:

Bad model	Good model
One enormous flat table	Star schema — fact + dimensions
Inconsistent values (Dr Smith vs D. Smith)	One row per consultant, unique ID
Dates stored as text	Proper DATE data types
80-line query, 12 minutes, uncertain results	10-line query, seconds, trusted results

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How to validate a data model generated by AI

AI can draft a data model. You still need to validate it.

Five-point validation checklist:

- Every table has a primary key
- Relationships and foreign keys are in the right place
- Data types match what the data actually is
- No duplication — each fact lives in one place
- Tested against real business questions

The mindset to reinforce:

AI output = a smart first draft. Your domain knowledge = what makes it correct.

What AI doesn't know:

- Your organisation's specific definitions
- Edge cases in your data
- Business rules that aren't written down

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